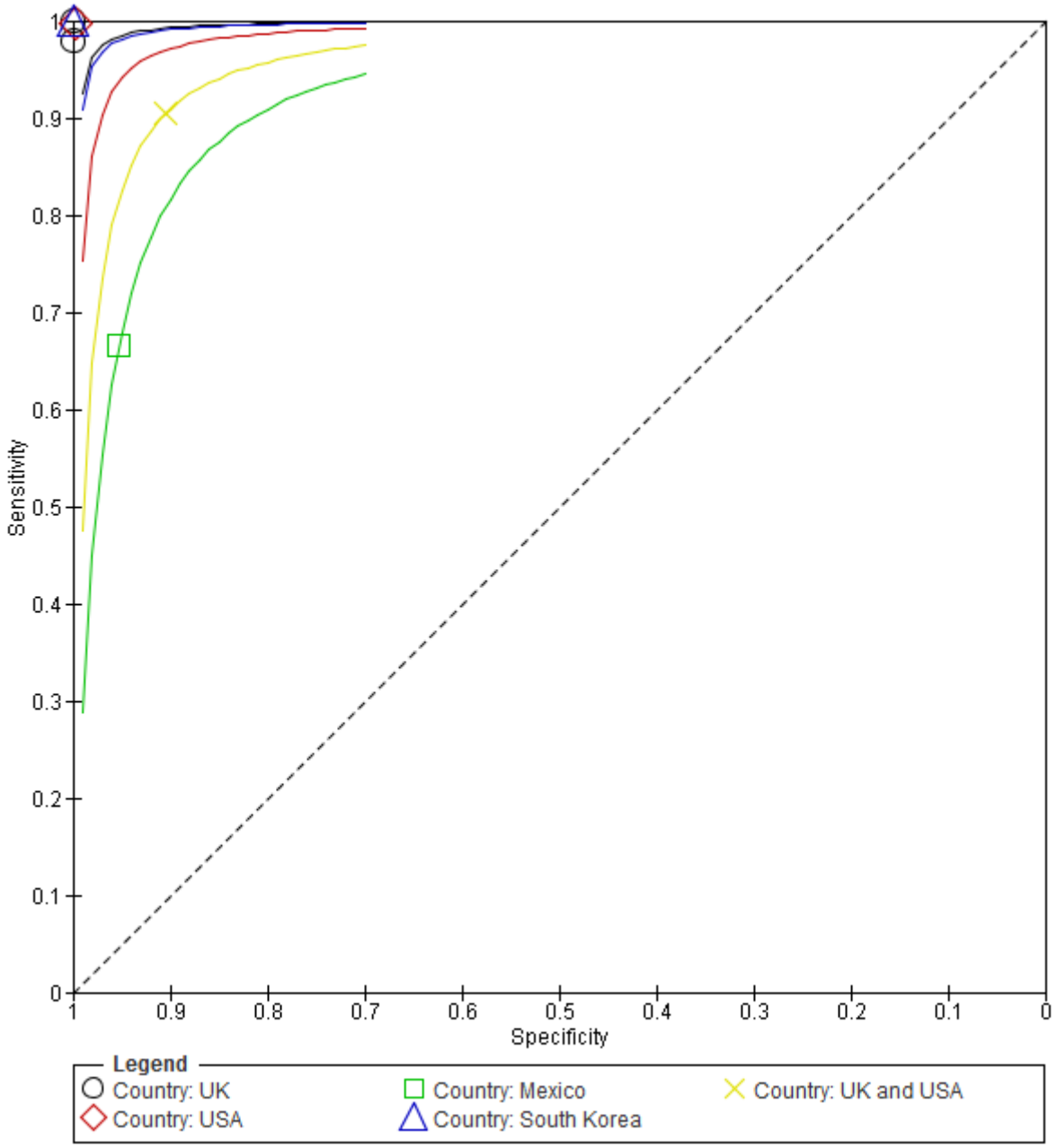
Supplementary

Supplementary 1

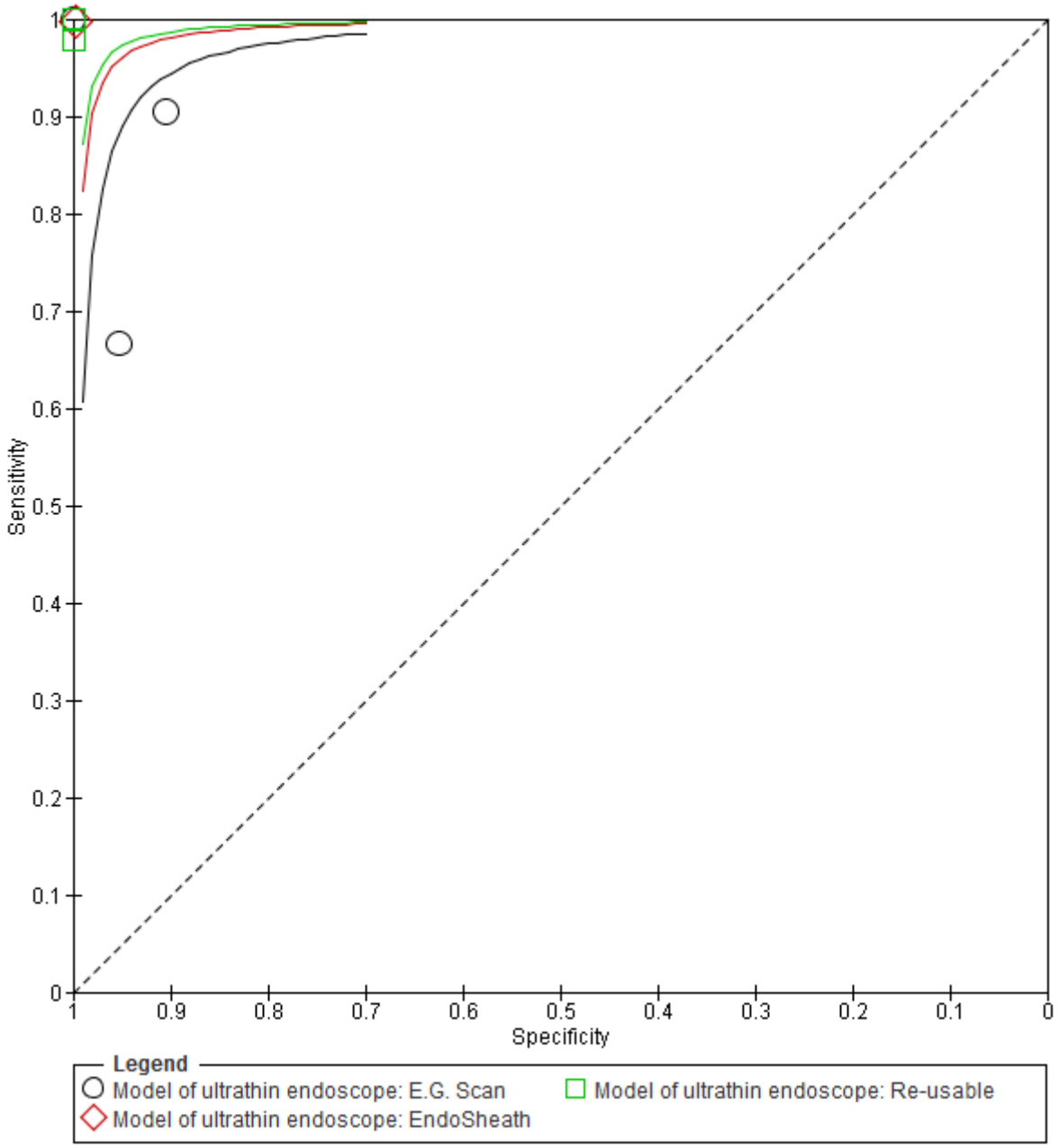
The following terms were combined as free text (title, abstract and if applicable keyword) and, if applicable, as Medical Subject Headings (MeSH-term) or Emtree: (Barrett Esophagus OR Barrett* OR oesophag* OR esophag* OR gastric OR stomach) AND (transnasal* OR trans-nasal* OR nasal* OR ultra-thin OR ultrathin OR small-caliber OR small caliber OR small-calibre OR small calibre OR super-thin OR superthin OR uTNE OR TNE OR oesophagoscop* OR esophagoscop*).

Supplementary 2. Assessment of patient tolerability, technical success rate, and safety of uTNE.

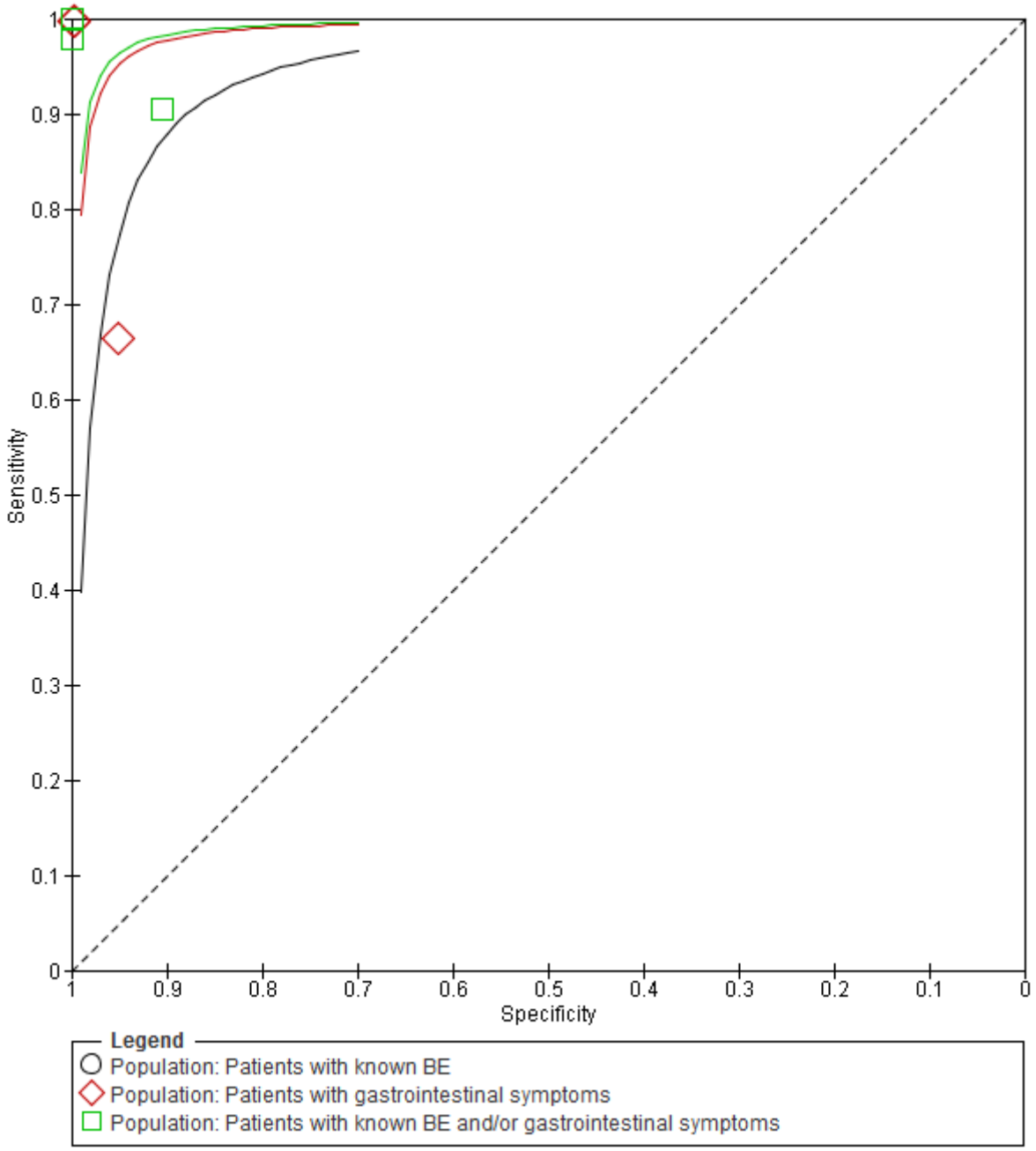
Author	Patient tolerability	Technical success rate %	Safety
Aedo et al. 2014[26]	Nasal introduction caused no or only mild pain in 77 of 96 patients (80%) and moderate pain in 19 patients (20%). The majority of patients did not experience nausea (88%), no comparison with cEGD	100	No serious adverse events
Sami et al. 2019[31]	Overall patient tolerability (VAS scores) was higher with E.G. Scan II compared with cEGD	89	5 non-serious adverse events: N = 2 probe technical failure N = 2 vasovagal symptoms after abdominal distention secondary to air insufflation during the procedure N = 1 self-limiting epistaxis
Kang et al. 2019[29]	Patients were more comfortable with ultrathin endoscopy (p=0.030) than with conventional endoscopy, with a lower symptom score for ultrathin endoscopy	100	Unknown
Shariff et al. 2016[33]	Significantly better experience with uTNE. 12 participants (60%) reported a preference for uTNE with local anaesthesia, 5 (25%) preferred cEGD with sedation and 3 (15%) had no preference	Unknown	Unknown
Catanzaro et al. 2003[27]	Tolerance of an unsedated examination as characterized by choking, pain, and overall discomfort was significantly better with the 4-mm endoscope when compared with historical data collected in their previous study of the 3.1 mm endoscope, no comparison with cEGD	96.1	1 non-serious adverse event: minor oozing of blood at the gastro-esophageal junction after a patient retched during an unsedated examination
Shariff et al. 2012[32]	No significant difference in VAS score for experience	95	No adverse events
Saeian et al. 2002[30]	No significant difference in VAS score for choking, sore throat, discomfort, and overall acceptability	100	No adverse events
Jobe et al. 2006[28]	No difference in patient tolerability, the majority of subjects chose uTNE over cEGD (71% [81/114] vs. 29% [33/114], respectively	98.3	6 non-serious adverse events: N = 1 gagging and pain N = 1 gagging N = 2 nonsyncopal vasovagal reaction secondary to rapid gastric insufflation N = 1 minor epistaxis



Supplementary 3. Subgroup analysis 'Country' in seven studies that reported the diagnostic accuracy of detecting columnar epithelium.

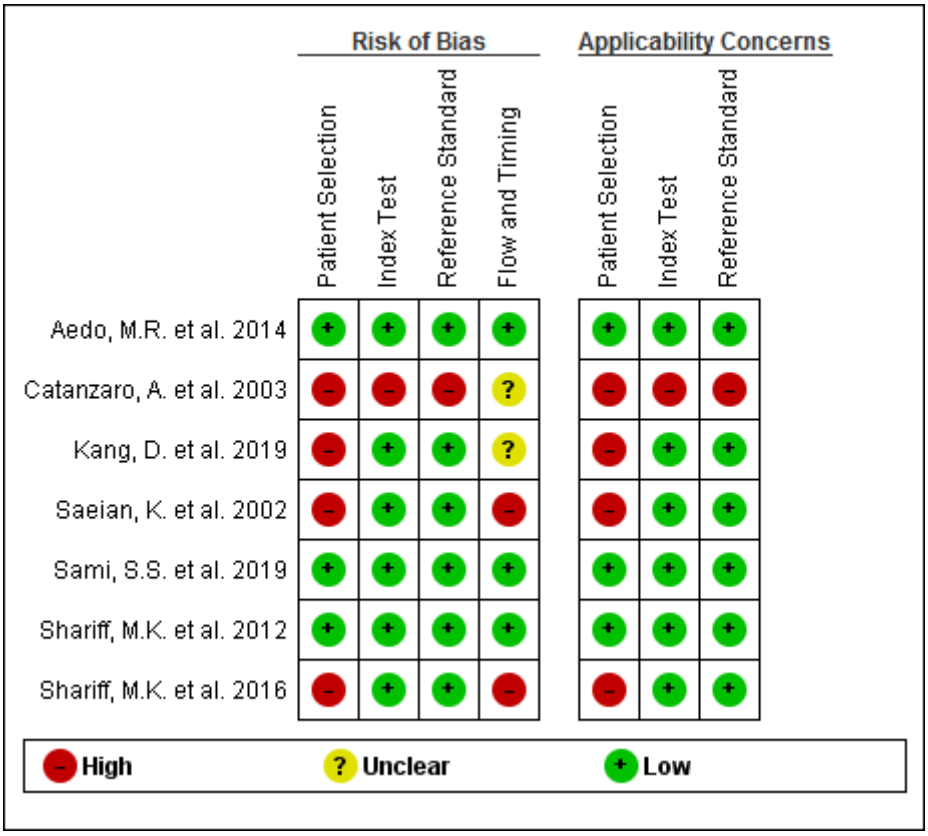


Supplementary 4. Subgroup analysis 'Model of ultrathin endoscope' in seven studies that reported the diagnostic accuracy of detecting columnar epithelium.

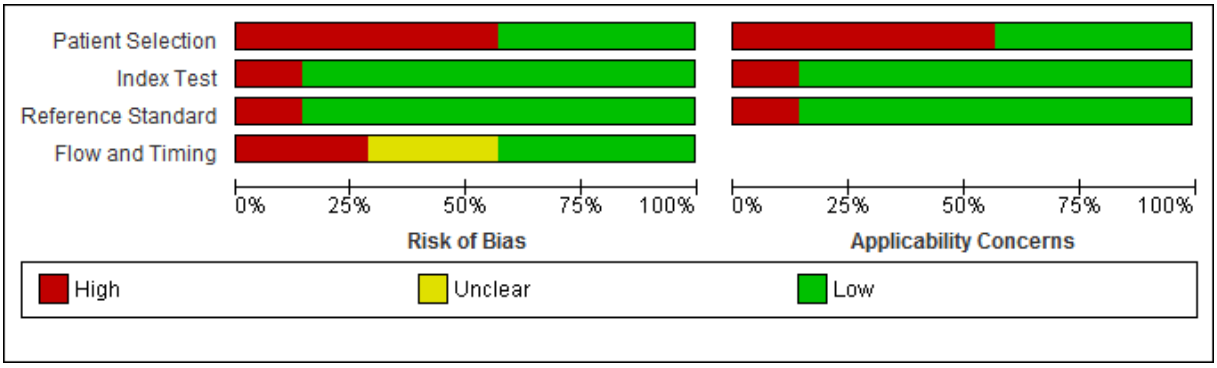


Supplementary 5. Subgroup analysis 'Population' in seven studies that reported the diagnostic accuracy of detecting columnar epithelium.

(A)

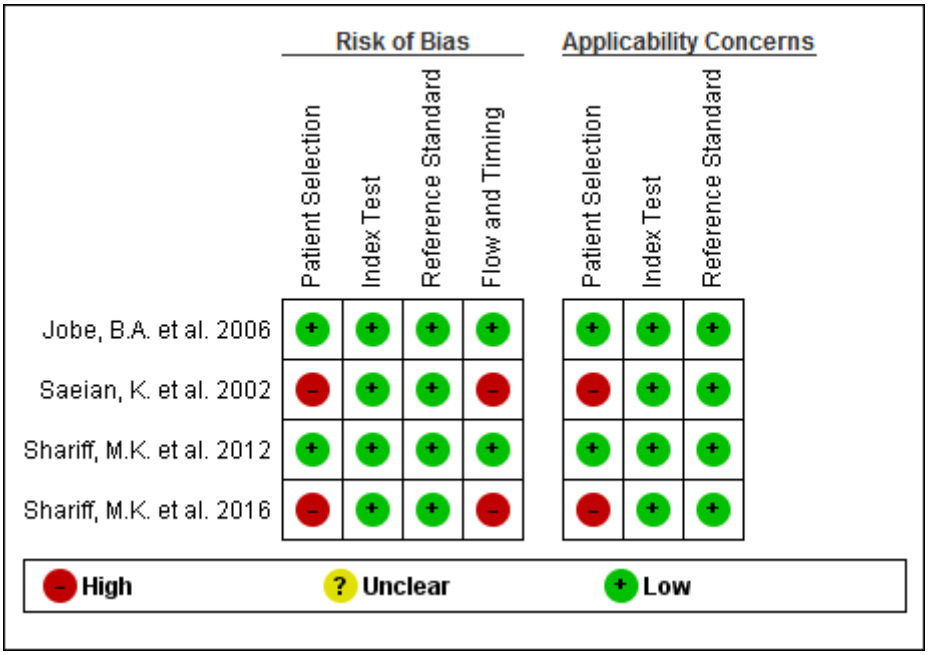


(B)

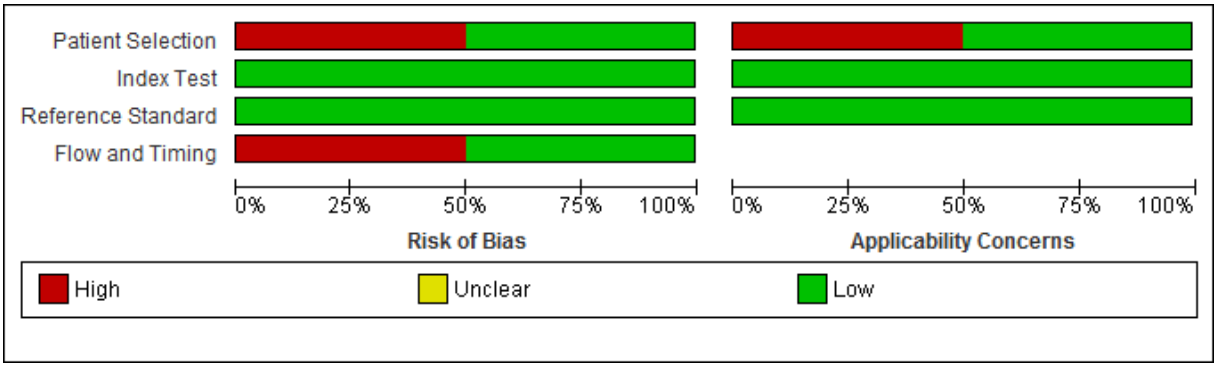


Supplementary 6. (A) The risk of bias and applicability concerns summary depicts risk of bias and applicability concerns ratings for each included study. (B) The risk of bias and applicability concerns graph is a cumulative bar plot across all included studies.

(A)



(B)



Supplementary 7. (A) The risk of bias and applicability concerns summary depicts risk of bias and applicability concerns ratings for each included study. (B) The risk of bias and applicability concerns graph is a cumulative bar plot across all included studies.